Lecture 2

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Today's Lecture

1. Reading Summary - Chapter 2
2. Observational Studies
   - Confounding
   - Cross Tabulation (Example: Smoking)
3. Natural Experiments
   - Regression Discontinuity Designs
   - John Snow on Cholera

Next Monday is Labor Day - no class.

Read Chapters 3-4 for next Wednesday.

Quiz 1 next Wednesday covers HW1.

Reading Summary

Chapter 2.1: Introduction

- Three notions of control
  - A control: Someone who doesn't receive treatment
  - Controlled experiment: A study where the investigator decides who gets treatment
  - Control for a confounder: Adjusting an analysis to account for confounding
- Cross Tabulation – Making more and more refined comparisons

Chapter 2.2: The Clofibrate Trial

- Watch Out: Even within an experiment, some comparisons are observational.
- Adherers are usually different in many ways than non-adherers.

Chapter 2.3: More Examples

- Several good examples. “Association is not causation.”

Chapter 2.4: Sex Bias in Graduate Admissions

- Simpson's paradox: "Relationships between percentages in subgroups can be reversed when the subgroups are combined."

Chapter 2.5: Confounding

- A confounder is a variable that is associated with both the outcome and the treatment.
- The confounder may explain associations between the outcome and the treatment.
- Any study design needs to consider possible confounders and control for them as much as possible.

Observational Studies

- In an **observational study**, the researcher **does not** decide who will get treatment and who will get control.
- Often the subjects decide themselves.
- Sometimes some other external factor decides.

Confounding

- Key problem with observational studies: confounding.
- **Confounder**: A variable that:
  1. is associated with treatment, and
  2. has an effect on response.
- Confounders are more of a problem with observational studies than with randomized controlled experiments
  - There isn't a mechanism to create similar treatment and control groups with observational studies.
- When confounders are present, it's hard to say whether any observed difference in response between the treatment group and the control group is due to treatment, or due to the confounders.
- Why would you ever do an observational study?

Example: Smoking

- Smokers have a higher rate of lung cancer.
- Does smoking cause lung cancer?
  - For a long time, people were unsure, because of possible confounders.
- Possible confounders:
  - Gender
  - Age
  - Occupation
  - Urban / rural
  - Alcohol
  - Genetics
  - And many more...

Cross Tabulation

- To deal with confounding, a researcher might make more and more refined comparisons.
- Example: Compare men who smoke to men who don't smoke; compare women who smoke to women who don't smoke.
- Another Example: Compare men who live in cities who smoke to men who live in cities who don't smoke. Also compare men who live in the countryside who smoke to men who live in the countryside who don't smoke. And similarly for women.
- This process is called **cross tabulation**.
  - Often the results are presented in a table.

Cross Tabulation: Two Main Problems

- There are two main problems with cross tabulation.
  1. Very quickly run out of data if you look at several confounders
  2. Always possible you forgot an important confounder
- The first problem can be solved with more data, or possibly with more advanced statistical techniques.
- The second problem is more fundamental, and generally can't be solved.

Discussion Question – How Much Data?

- Suppose you are doing a study on whether smoking causes diabetes, and want to control for age, represented by 5 categories:
  - Under 18
  - 18 - 35
  - 35.1 - 50
  - 50.1 - 60
  - 60+
- and sex, represented as male/female.
- What is the minimum number of people you need to complete the cross tabulation if you want each comparison to have at least 5 smokers and 5 non-smokers?

Natural Experiments

- Natural Experiments are somewhere in between experiments and observational studies.
- The researcher **does not** decide who gets treatment and who gets control.
- The subjects also **do not** decide whether they will get treatment or control.
- Instead, some external factor determines treatment.
- In some cases, the researcher can argue that the external factor is essentially random.
- Nature performs an experiment for us.

Example: Education and Income

- Research question: Does more education lead to higher income?
- More highly educated people do have higher income, but...
- Confounders:
  - Motivation
  - Parents' socio-economic status
  - Natural ability
  - Many others...

Regression Discontinuity Design

- In a regression discontinuity design, the assignment to treatment or control uses a strict, arbitrary cut-off for a score where the study participants have limited control.
- People just below and just above the cut-off may be similar. You should check this though.
- Example: In some countries, a score on a standardized test determines whether you go to college or a trade school.
  - You can compare people who just barely pass, and therefore attend college, to people who just barely don't pass, and therefore go to a trade school.
  - These two groups of people were in a sense randomly assigned to treatment (college) or control (trade school).

John Snow on Cholera

- https://en.wikipedia.org/wiki/John_Snow
- Most of the material in the next several slides comes from *Statistical Methods: Theory and Practice*, by David Freedman.

What causes Cholera?

- In the 1850s, there was considerable controversy.
- Germs had not yet been discovered, and were only one (unpopular) theory.
- The more popular theories were:
  - Miasmas (bad air)
  - Imbalances in the humors of the body (black bile, yellow bile, blood, etc.)
  - Poison in the ground

John Snow

- John Snow was a physician in London.
  - After extensive observation of Cholera cases, he came to believe believe that a living organism ingested in water or food caused Cholera.
  - The organism multiplies within the body, and escapes into the environment.
- John Snow developed arguments to support his theory, for example:
  - Cholera spreads along trading routes.
  - When sailors went somewhere with Cholera, they wouldn't get sick if no one left the boat.
  - These facts are hard to explain by the miasma theory, but easy to explain by the germ theory.

Epidemic of 1848

- John Snow identified the "index" (first) case in London:
  - a seaman named John Harnold.
  - He had newly arrived by the *Elbe* steamer from Hamburgh, where the disease was prevalent.
- Snow also identified the second case:
  - the man who took Harnold's room after Harnold died.
- Snow found several adjacent apartment buildings, with one hit by Cholera, and one not.
  - All of the Cholera-affected buildings had water contaminated by sewage, while the other buildings had relatively pure water.
- Pairing the adjacent buildings, presumed to be similar, is powerful statistically.

Epidemic of 1854

- Snow made a spot map of Cholera deaths.
  - The cases clustered around the Broad Street pump.
- There was a brewery near the pump, but none of the workers got sick.
  - The brewery had its own private pump on-site.
  - Most of the people drank beer anyway.
- A woman in Hampstead, outside of the map area, got Cholera.
  - She had water from the Broad Street pump delivered to her, because she liked the taste.
- All of this evidence is easy to explain by the germ theory, and hard to explain by the miasma theory.

The Cholera Epidemic: A Natural Experiment

- There were several water companies in London, and their service areas overlapped.
- On a single street, some people would have one company, others another company.
- The pipes had been laid many years before, when the water companies were still in active competition.
- In 1852, the Lambeth company moved its intake pipe upstream to get purer water.
- Snow compared the death rates between those who got water from Lambeth, and those who got water from Southwark and Vauxhall.

Comparison of Cholera Rates by Water Company

	Houses	Cholera Deaths	Rate per 10,000
Southwark and Vauxhall	40,046	1,263	**315**
Lambeth	26,107	98	**37**

Confounding?

- See Snow_quotes.pdf on Canvas
- Snow made a strong, convincing statistical argument, and made a major scientific advance.
- There was no fancy math, or complicated formulas. Instead, there was:
  - high quality data that he collected through considerable effort,
  - very careful use of the method of comparison, and
  - a strong argument linking the data, the assumptions, and the conclusion.
- See Review_questions1-2.pdf on Canvas for review questions from the first two lectures.